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Magnetic Field Induction and Doping Synergy Enable Efficient and Stable Perovskite Solar Cells

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Abstract

We present magnetic-field induced facile fabrication of highly crystalline, stable, and high efficiency formamidinium-lead-iodide perovskite solar cells. This work represents the first demonstration of computation-guided dopant screening combined with field-assisted perovskite crystallization. First-principles dopant screening identified neodymium as the optimal lattice-stabilizing candidate, guiding our subsequent experimental efforts. Synergistically coupling neodymium doping with field-assisted crystallization led to pronounced improvements in film quality, photovoltaic performance, and operational stability. Devices fabricated under magnetic field achieved a power conversion efficiency of ~21.1%, whereas those incorporating 0.08% neodymium reached a record efficiency 23.9%. The unencapsulated devices retained more than 95% of its initial efficiency after ~900 h of maximum power point tracking at room-temperature and 50% relative humidity under continuous simulated illumination. With emphasis on utilizing dopant-field synergy to boost device performance, our systematic and

effective approach, that bridges theoretical and experimental procedures, offers a rational design path toward high-efficiency, long-term-stable perovskite photovoltaics.

Keywords: perovskite solar cells, magnetic-field-assisted crystallization, transition metals, surface energy, formation energy.

1. Introduction

Perovskites, which are expected to significantly reduce the installation costs of solar power plants, have aroused great curiosity in the scientific community. Today's best-performing hybrid organic-inorganic perovskite (HOIP) solar cells have reached an excellent power conversion efficiency (PCE) exceeding 27% [1,2]. Despite this rapid advancement, further improvements in the chemical synthesis and manufacturing processes of perovskite films remain essential for enhancing device performance and durability [3-5]. Among the various strategies explored, composition engineering has emerged as an effective approach to optimize optoelectronic properties and suppress defect-induced recombination losses [6-9].

Among HOIPs, formamidinium lead iodide (FAPbI₃) material is one the most attractive absorber systems for perovskite photovoltaic applications because it combines a near-ideal bandgap, strong visible-light absorption, and excellent potential for high device efficiency. At the same time, its practical use remains challenging because the photoactive black perovskite phase requires rigorous control of crystallization, lattice stability, and defect formation. The features make FAPbI₃ a particularly relevant system for studying how dopants and external fields film formation and device performance [7].

In particular, modification of the B-site, traditionally occupied by lead (Pb²⁺), has gained increasing interest due to its direct influence on crystal structure, carrier dynamics, and environmental stability. Pb²⁺ in HOIPs induces intrinsic lattice instability due to its soft, relatively weak lead-halide bonding and stereochemically active 6s² lone pair. This results in a structurally labile framework with low formation

energy, making the material prone to relatively poor film formation, phase degradation and defect formation. Consequently, Pb-based perovskites suffer from limited long-term stability under operational conditions [10,11]. However, the fundamental effects of Pb^{2+} substitution with alternative metal cations are not yet fully understood. Recent studies have investigated the incorporation of transition metals such as iron (Fe) [12,13], cobalt (Co) [12,14], nickel (Ni) [12,15], manganese (Mn) [16] elements, widely used in magnetic materials, as dopants in perovskite solar cells (PSCs), revealing complex interactions between their electronic configurations and the perovskite lattice [17]. Klug et al. showed that partial substitution of Pb^{2+} in MAPbI_3 with transition-metal cations can effectively modify the lattice and improve device performance; notably, Co^{2+} doping yielded a PCE of 17.2% in HOIP [12]. Similarly, M. Abdi-Jalebi et al. reached a 21.6% efficiency with a low concentration of Rb^+ doping [18]. More recently, one of the most striking advancements in B-site doping is the incorporation of Nd^{3+} into FAPbI_3 [19], which led to a remarkable PCE of 23.68% due to improved phase stability and reduced non-radiative recombination [20].

In polycrystalline FAPbI_3 films, since photovoltaic performance is strongly coupled to crystal quality, parameters such as phase purity, lattice distortion, microstrain, preferred orientation, grain density directly affect the electronic defect density, carrier scattering, interfacial recombination, and charge-transport losses. Therefore, in general, a more ordered film with reduced microstrain and less defective grain boundaries is expected to suppress non-radiative recombination and improve performance. This connection is also important for FAPbI_3 films, where small changes in crystallization behavior can strongly influence both efficiency and long-term stability. From this viewpoint, dopant selection should not only be considered in terms of chemical incorporation, but also in terms of how the dopant leads to better stability, reduced defect formation, and field-responsive crystallization (see **Table S1**).

HOIP materials are sensitive to external stimuli due to their “soft” nature [21]. Simultaneously, nucleation rates and film texture can be influenced by external fields, potentially affecting crystal

orientation, as suggested in previous studies [22]. Wang et al. showed that solar cell efficiency could be increased from 16.72% to 18.56% by applying an 80 mT external magnetic field (MF) during the film growth [23]. In a similar work, Williams et al. performed film growth under an external MF by placing low concentration Fe on the Pb lattice ($\text{MAPb}_{0.9}\text{Fe}_{0.1}\text{I}_3$) and reported that MF increased the grain size and improved the film coverage [24]. However, the addition of Fe atoms seriously reduced the optoelectronic performance of HOIP films due to the destructive effects of Fe on the optoelectronic properties independent of the MF [25]. The results obtained by Williams et al. showed that competing effects can occur with the presence of an additive in conjunction with the application of external fields. At this stage, ferromagnetic materials, lanthanides or exchange transition metals can improve the structural and optoelectronic properties of FAPbI_3 crystals when grown under MF and may bring the efficiency of such solar cells closer to the theoretical limit. However, experimentally utilizing each of these candidate elements into the FAPbI_3 lattice can be extremely demanding, labor-intensive, and time-consuming. Therefore, performing a computational pre-screening of the candidate elements to identify those with the most favorable structural stability and optoelectronic properties can greatly reduce both experimental workload and cost [26,27].

In this study, we aimed to systematically screen a wide range of transition metal elements ($\text{X} = \text{Cr}, \text{Mn}, \text{Fe}, \text{Co}, \text{Ni}, \text{Zn}, \text{Cu}, \text{Nd}, \text{Ti}$ and Pt elements) using first-principles calculations to evaluate their feasibility as B-site dopants for the FAPbI_3 perovskite lattice under a MF. Our computational analysis focuses on dopant incorporation energies, lattice distortions, defect formation pathways, and electronic structure modifications. By identifying the most promising dopants and subsequently validating their effects experimentally, we seek to establish a rational framework that connects theoretical predictions with practical improvements in film quality, defect suppression, and photovoltaic performance.

The structures obtained from density functional theory (DFT) and magnetization calculations revealed that X-doped FAPbI_3 (X.FAPbI_3) exhibits higher magnetization and increased magnetic susceptibility

compared to pristine FAPbI₃, indicating that suitable dopants can amplify MF-lattice interactions during crystallization. Electronic density-of-states (DOS) analyses further demonstrated that only a subset of dopants, most notably Nd³⁺, suppresses trap states across substitutional (X_{FA}, X_i, X_{Pb}) configurations, a critical requirement for maintaining high carrier mobility and minimizing non-radiative recombination in the final device. Additionally, surface-energy calculations for the various terminations of X.FAPbI₃ showed that Nd³⁺ most effectively reduces surface energies, suggesting enhanced structural stability. By combining these computational insights, our screening strategy enabled the identification of Nd³⁺ as the most promising dopant for producing high-quality films and high-performance photovoltaic devices under MF-assisted crystallization. This rational selection forms the basis for the experimental component of our study, which connects theoretical predictions with measurable improvements in film microstructure, defect suppression, and overall device efficiency.

Based on these computational insights, we selected Nd³⁺ as the optimal dopant for further experimental validation under MF-assisted crystallization. This study therefore couples first-principles dopant screening with MF-guided film growth to reveal a synergistic mechanism for structural ordering and defect suppression in FAPbI₃. Unlike previous studies that investigated either Nd³⁺ doping or MF-assisted MAPbI₃ growth separately, our work demonstrates a synergistic enhancement where the incorporation of Nd³⁺ amplifies the impact of the MF on crystallization, leading to superior structural order, enhanced charge transport, and improved device stability. To the best of our knowledge, the growth and crystallization behavior of both pristine and Nd³⁺-doped FAPbI₃ under an external MF have not been previously reported (see **Table S1**). Our findings indicate that perovskite films grown under an MF develop a well-defined magnetic easy axis, reflecting a high degree of crystallographic ordering along this direction.

We therefore fabricated control and Nd.FAPbI₃ devices with and without an external MF and evaluated their photovoltaic properties. Remarkably, devices incorporating 0.08% Nd³⁺ and growing under a MF

achieved a record PCE of 23.9%, demonstrating the strong synergistic coupling between Nd^{3+} doping and MF-directed crystallization. Beyond efficiency enhancement, MF treatment significantly improved operational stability; while undoped control devices retained only ~21% of their initial PCE after 300 h without an MF, applying an MF increased PCE retention to ~68% over the same period. Even more notably, the unencapsulated Nd^{3+} -doped PSC maintained over 95% of its initial PCE after ~900 h of maximum power point tracking (MPPT) under continuous illumination. These results highlight the effectiveness of combining targeted B-site doping with MF-assisted crystallization as a powerful dual-engineering strategy for achieving highly efficient and stable perovskite photovoltaics.

2. Results and Discussion

In the sections that follow, we detail the computational screening workflow, the MF-assisted fabrication method, the structural evolution of the films, and the resulting device performance. Such findings suggest that MF can act as an additional degree of freedom for tuning microstructure. Transition-metal dopants, which inherently modify the magnetic susceptibility of the perovskite lattice, may therefore further amplify the effect of MF-assisted crystallization, offering a promising route to modulate both structural and optoelectronic characteristics.

2.1. Dopant screening via computations

First, we performed electron spin resonance (ESR) measurements, as shown in **Figure 1A,B** and **Figure S1**, to investigate the influence of the MF on the perovskite films. The observed resonance and its angular variation indicate that MF-treated control (*i.e.*, control@MF) film is magnetically anisotropic and exhibits a preferred magnetic easy axis, rather than a fully isotropic response. We interpret this ESR behavior as evidence that the applied MF influences local spin environment, most likely associated with

unpaired-spin states in the Pb-I framework, promoting a more directional structural ordering during crystallization. Thus, the ESR results confirmed that the perovskite structure is indeed responsive to magnetization, motivating further investigation into how B-site doping could also enhance the device quality. Based on this, we conducted extensive computational screening to identify promising dopants that could *synergistically* improve the material's crystallinity and device efficiency under MF.

As shown in **Figure 1C**, the screening procedure to achieve high PCE in our study consists of three main steps: (i) pre-screening; we selected 10 candidate elements out of the periodic table based on their elemental magnetic moments and valency (ii) computational-screening; we performed extensive DFT-guided screening calculations, over FAPbI₃ perovskite containing these 10 candidate dopants, based on selected descriptors such as formation energy, DOS and surface energy, and (iii) device measurements: the best candidate dopant is selected for MF-treated film fabrication and further device studies for validation (**Figure S2-S10** and **Table S2**).

As shown in **Figure 2A** (see also **Figure S2**), we identified that the dopants can occupy three distinct positions within the perovskite lattice: X_{FA} (where the dopant substitutes the FA site), X_i (where the dopant occupies an interstitial site), and X_{Pb} (where the dopant substitutes the Pb site). The lattice formation energies associated with these positions help inform which dopants are more likely to stabilize the structure, especially under external MF conditions. For instance, Nd³⁺ in the X_i (interstitial) position, as shown in **Figure S2**, induces a more significant lattice expansion and less strain compared to other dopants, a behavior observed experimentally as well. **Figure 2B** illustrates that doping significantly enhances the magnetic susceptibility of FAPbI₃, making the material more responsive to the applied MF. This enhanced magnetization is essential because it directly correlates with the improvement in the alignment and growth of perovskite crystals under the influence of MF. The MF-assisted crystallization, in turn, leads to a more ordered and stable crystalline structure, which is critical for minimizing defect formation and enhancing the charge transport properties of the perovskite film.

Further exploration of the effects of dopant incorporation under MF conditions on electrical and structural stability is shown in **Figure 2**. **Figure 2C** demonstrates the DFT-calculated formation energies of various doped FAPbI₃ structures, both with and without an MF. The results clearly show that the doped perovskite can be stabilized by the external MF, indicated by the decreased formation energies. This is a critical point for device performance, as lower formation energies suggest that the material can more easily form stable, high-quality crystals under MF, reducing the likelihood of defects that would otherwise hinder the efficiency and stability of the PSCs. A comparison of the doped and undoped perovskites is shown by the Density of States (DOS) graphs in **Figure S7**. Few dopants provide a proper trap-free state for X_{FA}, X_i, X_{Pb} configurations (Nd³⁺ and Co²⁺ specifically) in particular (**Figure 2E**), which can affect carrier lifetimes and recombination rates. To maintain high carrier mobility and reduce non-radiative recombination losses in the completed device, it is essential to minimize trap density via controlled doping and MF application. Surface energies calculation of doped X.FAPbI₃ for different terminations (**Figure 2D**) pointed out that Nd³⁺ reduced the surface energies the most, suggesting a further perovskite stabilization. Thus, we have concluded that Nd³⁺ doping is the most logical candidate for high-quality film growth and high-performance device measurements under MFs (**Figure 2F**).

Taken together, **Figure 2** identifies Nd³⁺ as the most promising dopant not from a single calculated property, but rather from the best overall balance of MF responsiveness, electronic purity, and structural stabilization. Specifically, Nd.FAPbI₃ shows enhanced magnetic susceptibility, favorable formation under MF, reduced trap-state character in the DOS analysis, and the lowest average surface energy among screened candidates. These predictions guided our experimental selection of Nd³⁺ for MF-assisted film growth. Consistent with our screening results, we observed that the Nd-doped films prepared under MF exhibit lower-angle XRD shift and reduced microstrain, stronger PL, longer carrier lifetime, lower trap density, improved recombination resistance, more compact morphology, higher device efficiency, and markedly better operational stability.

2.2. Preparation and characterization of the perovskite films

The perovskite absorber layers, which is doped with Nd ion determined as a result of DFT-based dopant screening calculations, is coated using the solution-based spin coating method and a steady MF is applied via a permanent magnet during annealing (**Figure 3A**). Here, 0.08% Nd refers to the nominal molar ratio of $\text{Nd}(\text{CF}_3\text{SO}_3)_3$ relative to Pb in the PbI_2 precursor solution used for film preparation, and this nominal composition was used in all subsequent film and device characterizations (see **Figure S11**). The concentration-dependent device optimization identifies 0.08% Nd as the optimum nominal loading (**Figure S11**). Although all tested Nd-containing devices outperform the undoped control, increasing the nominal Nd loading beyond 0.08% does not provide further improvement, suggesting that a trace amount is sufficient to realize the beneficial effect. XPS characterization of the nominally 0.08% Nd-doped film further confirms the presence of Nd in the final perovskite film (**Figure S12**). The application of an MF induces lattice distortions, causing the X-ray diffraction (XRD) peak positions to shift to higher angles in control samples, which indicates lattice contraction (**Figure 3B**). In contrast, Nd.FAPbI_3 films prepared under a MF exhibit a slight shift in XRD peak positions to lower angles, suggesting lattice expansion. The lower-angle shift is therefore consistent with Nd incorporation into the perovskite lattice. Together with the reduced micro strain from Williamson–Hall analysis and the DFT comparison of possible Nd incorporation sites, these observations suggest that Nd incorporation most likely has a predominant interstitial character. However, as the present experimental evidence does not unambiguously distinguish interstitial from substitutional occupancy, substitutional incorporation cannot be fully ruled out. In particular, the Nd^{3+} -doped film shows a lower micro strain value (1.42×10^{-4}) than both the undoped (2.83×10^{-4}) and control@MF films (4.05×10^{-4}), indicating that Nd^{3+} doping alleviates MF-induced lattice stress while promoting structural expansion (Table S3). Our structural interpretation is based primarily on the systematic shift of the principal α - FAPbI_3 diffraction peak near 14° , which is clearly resolved and

directly related to the perovskite lattice. By contrast, the weak residual PbI_2 -related feature near 12.7° was not used as the basis of the lattice analysis, and any small apparent variation in this region is not interpreted here as a meaningful structural shift of the PbI_2 phase. To further clarify how the MF regulates the perovskite crystallization process, we analyzed the XRD patterns of FAPbI_3 films prepared with and without MF treatment. As shown in **Figure 3B** and **Figure S13**, the film annealed under an MF exhibits a significantly sharper (100) diffraction peak centered at 14.28° with a full width at half maximum (FWHM) of 0.108° , compared to the same peak in the control sample (no field), which appears at 14.19° with a broader FWHM of 0.173° . This narrowing of the diffraction peak confirms improved crystallinity and enhanced structural coherence in the field-treated sample. This enhanced crystallinity under MF is consistent with the DFT-predicted reduction in formation energy and surface energy for Nd-doped FAPbI_3 , indicating that MF-assisted crystallization energetically favors the α -phase formation. These results support a mechanistic picture in which the MF influences crystallization by biasing the organization of dipolar precursor species during annealing. In the presence of paramagnetic Nd^{3+} dopants, this effect is expected to become stronger because Nd incorporation enhances the magnetic response of the perovskite system, as indicated by the DFT calculations. This interpretation is also supported by the ESR indication that MF-treated perovskite films are magnetically responsive, together with the XRD, microstrain, morphology, and carrier-lifetime results obtained for the Nd^{3+} -doped film prepared under MF. As a result, MF-assisted growth can suppress random nucleation, promote more directional crystal growth, and enhance long-range structural ordering. Although direct in situ evidence for this process is beyond the scope of the present study, the combined theoretical and experimental results support this interpretation.

Both the MF and Nd^{3+} doping induces changes in lattice strain, as analyzed using the Williamson–Hall method based on the broadening of XRD peaks (**Figure S14**). The slope of the Williamson–Hall plot ($\beta \cos\theta$ vs. $4\sin\theta$) increases from 2.83×10^{-4} for undoped FAPbI_3 to 4.05×10^{-4} for the control@MF,

indicating increased microstrain. However, in the Nd.FAPbI₃ film with MF, the slope decreases to 1.42×10^{-4} , suggesting reduced overall microstrain. This result, combined with the shift of the XRD peak to lower angles, suggests that interstitial incorporation of Nd³⁺ ions promotes lattice expansion while relieving MF-induced strain.

To gain insights into carrier dynamics and investigate the band gap, steady-state photoluminescence (PL) measurements were performed (**Figure 3C**). The control@MF sample shows lower PL intensity compared to the control. The PL intensity of the Nd-doped perovskite film prepared with a MF shows a noticeable enhancement and a small variation in the PL peak position compared to the control films. 0.08% Nd³⁺ doped sample prepared with MF significantly enhances the PL intensity, indicating that Nd doping with MF reduced trap densities, leading to reduced non-radiative recombination centers, improved radiative recombination and crystalline quality. All samples exhibit a PL peak around the same wavelength (~805 nm), indicating that neither the MF nor the Nd doping with MF significantly shifts the emission peak. This suggests that the bandgap of the material is not greatly affected by these modifications, and the observed changes in intensity are primarily due to improvements in carrier dynamics (i.e., reduced traps, better film quality). We further investigated the charge recombination kinetics in the corresponding devices by time-resolved photoluminescence (TRPL) (**Figure 3D**). The related data extracted from TRPL were listed in **Table S4**. The fast decay component τ_1 is attributed to the surface recombination and trap-assisted recombination at the grain boundaries, and the slow decay component τ_2 is related to radiative recombination inside the bulk of the perovskite films [28]. The τ_2 lifetime of the control sample (412.72 ns) was slightly prolonged to 515.05 ns for the control@MF sample, and significantly increased to 1322.02 ns for the Nd³⁺ doped sample with MF. Unlike prior Nd³⁺ doped films, which faced limitations in charge carrier dynamics, our results show that MF-assisted doping extends carrier lifetime (i.e., to 1322.02 ns), marking a significant improvement over reported values in the literature.

The fast decay component (τ_1) reduces from 95 ns to 52.55 ns for the control@MF sample and further to 39.65 ns for the Nd³⁺-doped sample with MF. This decreasing trend in τ_1 is attributed to more efficient interfacial charge transfer, rather than increased non-radiative losses.

To better understand TRPL results, we performed space charge-limited current (SCLC) measurements to probe defects in an alternative way. Trap density was experimentally estimated using SCLC measurements (**Figure S15**), confirming a reduction in trap states consistent with our DFT-guided predictions. Hole-only devices utilizing the structure of FTO/NiO_x/perovskite (control and Nd.FAPbI₃ with and without MF)/SpiroO-MeTAD/Au were fabricated and characterized by current-voltage (I-V) characteristics in dark conditions. The dark I-V curves reveal that trap-filled limit voltage (V_{TFL}) corresponds to trap densities of $6.98 \times 10^{15} \text{ cm}^{-3}$ for the control device, and $4.42 \times 10^{15} \text{ cm}^{-3}$ for the Nd.FAPbI₃ device with an MF. We can further analyze the correlation between the trap densities and the fast decay component τ_1 from the TRPL results. The control sample has high trap density, and it correlates with the long τ_1 , indicating that despite the high trap density, these traps might be deeper or less efficient at capturing carriers immediately after excitation. The trap density of Nd.FAPbI₃ prepared with an MF is slightly lower than that of the control@MF sample, yet it still results in a shorter τ_1 . This indicates that while the MF may passivate some shallow traps, reducing their contribution to non-radiative recombination, the overall trap states still dominate, leading to faster recombination dynamics during the early decay stage. The relation between trap density and the fast decay component τ_1 suggests that trap depth nature (shallow vs. deep) is as important as trap density. While the Nd³⁺ doping reduces the trap density, it introduces more shallow traps, leading to faster non-radiative recombination. The application of the MF appears to mitigate this effect by improving the film's overall quality via shallow trap passivation, even though the trap density slightly increases.

To understand the device behavior for τ_1 , we performed further TRPL measurements to probe the interaction between the perovskite active layer and the spiro-OMeTAD layers carefully. We study the impact of MF and Nd^{3+} doping on the charge transfer between the perovskite and HTL. At the interface between the perovskite and HTL, a built-in energy level difference is created due to the difference in the energy levels. This energy difference drives the photo carriers from the perovskite to the charge transport layer. τ_1 refers to the rapid transfer of charge carriers from perovskite to the transport layers, and τ_1 will be shorter (**Figure S16**). The τ_1 values of the control/HTL, control@MF/HTL and Nd.FAPbI_3 (with MF)/HTL samples are 14.11 ns, 23.91 ns, and 8.9 ns, respectively. Compared to the control perovskite/HTL samples, the Nd-doped perovskite (at MF)/HTL sample exhibits shorter τ_1 values, suggesting less efficient charge transfer at the control/HTL interface.

The influence of MF and Nd^{3+} doping on charge-transfer dynamics and the electrical properties of interfaces was further investigated through electrochemical impedance spectroscopy (EIS) measurements. EIS data were collected at voltage values near the open-circuit voltage (V_{oc}) under dark conditions, and the curves were fitted using the equivalent electrical circuit (**Figure S17**). From there, the series resistance (R_s), recombination resistance (R_{rec}) and capacitance were extracted (**Table S5**). The control sample and Nd.FAPbI_3 prepared with an MF exhibits larger R_{rec} values, indicating suppressed carrier recombination. The larger arc in the Nyquist plot corresponds to a higher R_{rec} , suggesting improved charge extraction and reduced charge recombination at the interfaces. Notably, the Nd.FAPbI_3 device prepared with MF shows higher R_{rec} values (2.888 k Ω) compared to the control device with a R_{rec} of 1.098 k Ω . This indicates more effective inhibition of charge recombination and enhanced carrier extraction, contributing to improved photovoltaic performance, which is consistent with the TRPL data. The application of an MF with Nd doping significantly reduces R_s . This reduction in R_s (from 7.921 to 5.534 Ω) is likely due to a combination of factors, including enhanced crystallinity, reduced trap density,

and possibly improved interfacial connectivity. These improvements collectively contribute to the observed increases in V_{OC} , FF, and PCE [29].

Higher R_{rec} typically suggests a more favorable surface morphology for efficient charge transport and reduced recombination. To better understand the enhanced R_{rec} in PSCs with the influence of an MF, scanning electron microscope (SEM) images of different perovskite films were obtained (**Figure 3E**). The control sample has an average grain size of 1.5 μm , while the control@MF film exhibits larger grain sizes (average of 2.3 μm) and no pinholes. This improvement correlates with the enhanced R_{rec} and reduced trap states. The control sample exhibits larger, irregular grain boundaries and crystalline grains with pinholes, which may create trap sites that could contribute to the lower R_{rec} observed in the EIS data. SEM micrographs show compact grains with well-defined grain boundaries and no pinholes for the Nd^{3+} -doped films with the MF (**Figure 3E**). The more compact and pinhole-free morphology observed by SEM agrees with the DFT-predicted decrease in surface energy for Nd-doped FAPbI_3 under MF, suggesting that the MF-induced ordering promotes energetically favorable surface reconstruction. The Nd.FAPbI_3 prepared with an MF shows an average grain size of 1.29 μm .

This indicates that although Nd^{3+} doping slightly reduces the grain size, it simultaneously improves the distribution of trap states by suppressing deep-level traps, thereby achieving an effective balance between microstructural and electronic properties. While the grain size is slightly reduced in the Nd^{3+} -doped sample under MF compared to the control, we observe a significant increase in grain size relative to Nd^{3+} -doped films grown without MF (not shown). This supports the conclusion that MF application enhances the grain growth dynamics in doped perovskite systems. The Nd.FAPbI_3 films prepared under an MF exhibit improved grain uniformity and reduced trap density, which help mitigate non-radiative recombination and improve photovoltaic performance. Atomic force microscopy (AFM) combined with Kelvin probe force microscopy (KPFM) was further applied to probe the effect of MF on the surface morphology and surface potential (**Figure S18**). While the overall topographic features do not show a

pronounced systematic change, the KPFM results reveal clear shifts in the contact potential difference (CPD). Compared with reference film, the control@MF sample shows a slightly shifted surface-potential distribution, whereas the Nd^{3+} -doped film prepared under MF exhibits a further shift (approximately -10.3 mV), indicating modified surface electronic properties and charge redistribution at the film surface. To further understand the influence of MF strength on device performance, we fabricated Nd.FAPbI_3 films under different MF intensities (0, ~60, ~100, and ~130 mT). As shown in **Figure S19**, the PCE peaks at 100 mT (~23.5%) and decreases at lower and higher field intensities. This non-monotonic trend indicates that magnetic-field-assisted crystallization operates within an optimum processing window, with the best device performance obtained at ~100 mT. At relatively low field strength (~60 mT), the magnetic interaction is likely insufficient to effectively bias precursor organization during annealing, even in the presence of paramagnetic Nd^{3+} dopants. At ~100 mT, this interaction becomes strong enough to promote more favorable crystallization, consistent with the improved properties achieved under the optimized condition, including the lowest microstrain, prolonged carrier lifetime, and enhanced recombination resistance for the Nd^{3+} -doped film prepared under MF. At higher field strengths, however, excessively strong field-induced organization may disturb nucleation homogeneity, for example through local over-alignment or partial clustering of polar or paramagnetic species, thereby partially offsetting the beneficial effect of MF-assisted crystallization. Although direct film characterization at each field strength or in situ evidence for this process is beyond the scope of the present study, the combined device optimization, DFT, and optimized-condition film characterization results support this interpretation.

2.3. The photovoltaic performance of perovskite solar cells

Planar-type PSCs with the structure of $\text{FTO/c-SnO}_2/\text{FAPbI}_3$ (control) and $\text{Nd.FAPbI}_3/\text{Spiro-OMeTAD}/\text{Au}$ were designed and fabricated, with and without the application of an MF. The J-V curves of the best performing control and Nd.FAPbI_3 devices prepared with MF under standard illumination at

100 mW cm⁻² are shown in **Figure 4A**. JSC remains almost the same, while V_{OC} substantially increases compared with the control device. The control@MF device suffers from severe hysteresis, while the hysteresis behavior in Nd.FAPbI₃ PSCs with MF were suppressed. The hysteresis index is reduced from 0.2 to 0.014, which could be related to enhanced charge transport and reduced trap-state density. Hysteresis is often caused by the movement of mobile ions within the perovskite layer, which can interfere with device performance. The presence of the MF and Nd³⁺ doping likely stabilizes ion movement and enhances charge carrier dynamics, resulting in more stable and reliable J-V characteristics with reduced hysteresis.

The steady-state power output (SPO) measurements at a fixed bias near the maximum power point (MPP) were carried out to assess the performance of PSCs. A stabilized efficiency of 20.66% and 23.48% was achieved for control and Nd³⁺ doped perovskite devices prepared with MF, respectively (**Figure 4B** and **Figure S20**). The stabilized PCE improved from 20.35% (control) to 20.66% with MF treatment alone and further increased to 23.48% upon synergistic application of Nd³⁺ doping and MF. Besides, we observed a stabilized PCE of 22.73% with Nd³⁺ doping only (i.e., no MF) (**Figure S20**), which led us to conclude that both the addition of Nd³⁺ and the application of an external MF collectively increase the device performance. Thus, a dual-modification approach, achieving a record PCE of ~24% and surpassing the performance of either Nd³⁺ doping alone or MF application alone, demonstrates the necessity of their combined effect for optimal device performance.

The external quantum efficiency (EQE) spectra of PSCs are compared in **Figure 4C**. As compared with PSCs by the control device and the Nd.FAPbI₃ prepared with the MF reveals enhancement in the wavelength (λ) ranging from 380 to 800 nm. The integrated J_{SC} values for the control perovskite devices prepared without and with MF and Nd³⁺ doped with MF are 24.27, 24.44, and 24.85 mA cm⁻², respectively. These J_{SC} values are in good agreement with the value measured from the J-V scan. All

devices exhibit a standard deviation of $< 5\%$. It is apparent that the devices prepared utilizing MF show a higher EQE, which is due to the less optical loss ($\sim 1\%$). **Figure 4D** shows the PCE histogram for devices with and without MF. Amazingly, the devices prepared with MF exhibit excellent repeatability with a very small standard deviation in contrast to the devices prepared without MF, confirming better reproducibility and affirming the observed enhancement in PCE.

Given the enhanced optoelectronic properties, extended carrier lifetime, and improved EQE and PCE values, combining an MF and Nd^{3+} doping is expected to create a more efficient active absorber layer in PSCs compared to FAPbI_3 alone. Device performance testing showed that the samples prepared under an MF achieved the highest PCE values. We assessed the photovoltaic performance of control and 0.08% Nd^{3+} -doped FAPbI_3 devices with an MF. The average PCE value increased from $20.22 \pm 0.42\%$ to $20.71 \pm 0.26\%$ with MF. Upon doping with Nd^{3+} and applying an MF, the PCE values for devices rise to $23.62 \pm 0.20\%$. It is worth noting that the highest PCE recorded among all devices was 23.89% (as reported in **Table S6**), while the average PCE for the Nd^{3+} -doped FAPbI_3 devices with MF treatment was $23.62 \pm 0.20\%$, as presented in **Figure 4D**. The key photovoltaic parameters, including J_{SC} , V_{OC} , FF, and PCE, are presented as histograms in **Figure 4E** (see **Figure S21**). Films prepared with MF exhibited good reproducibility. The control device gives an average $J_{\text{SC}} = 25.32 \text{ mA cm}^{-2}$, $V_{\text{OC}} = 1.048 \text{ V}$, and FF = 0.76. The devices with Nd^{3+} doped perovskite prepared with MF, the average J_{SC} , V_{OC} and FF are increased to 25.84 mA cm^{-2} , 1.103 V and 0.82, yielding an average PCE of 23.62% . The control perovskite cells fabricated with MF have an average V_{OC} of 1.058 V , J_{SC} of 25.43 mA cm^{-2} , and FF of 0.77. The overall effect of MF with doping indicates the key role of these approaches in enhancing the overall device efficiency and performance characteristics. Simultaneously, external MF and Nd^{3+} doping are very beneficial for device performance and reproducibility.

The long charge carrier diffusion length from TRPL (**Figure 3D**) would ensure that charge carriers are efficiently transported from the perovskite thin film to the electrodes, resulting in high J_{SC} for PSCs. Thus, PSCs by the doped and undoped FAPbI₃ devices prepared with MF exhibit enhanced J_{SC} , boosting PCE. The reduction in R_s along with higher R_{rec} from EIS helps improve the FF, leading to overall better device efficiency.

To further elucidate the mechanism of improvement of V_{OC} and FF, capacitance-voltage ($C-V$) measurements were performed to obtain information about the built-in potential (V_{bi}) in control and Nd.FAPbI₃ PSCs. As shown in **Figure S22**, the V_{bi} for the Nd.FAPbI₃ PSC is 1.046 V, higher than the control PSC ($V_{bi} = 0.984$ V). The relatively larger V_{bi} can be related to the stronger driving force to separate the photogenerated charge carriers, which points to the higher V_{OC} of the Nd.FAPbI₃ PSCs.

To investigate the operational stability of the devices, we exposed the unencapsulated devices to continuous light illumination using a white LED lamp under ambient conditions (25 ± 2 °C and $50 \pm 5\%$ RH) for 900 h. Nd-doped device prepared with an MF showed an efficiency drop of less than 5% after 900 hours, while the control@MF device decreased to 68% of its initial efficiency under the same conditions. (**Figure 4F**). This exceptional stability most likely arises from the combined effects of lattice stabilization, reduced defect density, and improved microstructural integrity under the MF–dopant synergy, which together help mitigate intrinsic degradation pathways in both the perovskite bulk and the relevant charge-transfer interfaces. In this picture, lattice stabilization and defect suppression appear to be the dominant contributions, while the more compact and pinhole-free film formation observed by SEM may provide an additional benefit by improving film continuity and reducing microstructural weak points. The PCE of the control sample decreases significantly over time, dropping to < 21% of its initial value after only 300 h, demonstrating poor stability due to the undesirable ion migrations under light soaking. Notably, MF application alone also significantly enhances the operational stability of undoped control devices, increasing PCE retention from ~21% (no field) to ~68% (with field) after 900 hours.

This suggests that the MF effectively stabilizes the perovskite layer through defect passivation and suppressed ion migration. Thus, the stability enhancement observed in our study, 95% PCE retention over 900 hours, is among the highest reported for unencapsulated PSCs, directly linked to suppressed ion migration and improved interfacial charge transport.

3. Conclusion

In summary, this work reports for the first time the MF-assisted growth of both pristine and rare-earth Nd^{3+} -doped FAPbI_3 perovskite films, achieving remarkable improvements in crystallinity, optoelectronic quality, and long-term device stability. The integration of a steady MF during the crystallization of Nd-doped perovskite films promoted enhanced crystal alignment, reduced trap density, and improved film morphology. These structural optimizations, in turn, facilitated superior charge carrier dynamics, as evidenced by longer carrier lifetimes and suppressed non-radiative recombination. Theoretical calculations predicted a reduction in formation energy, enhanced magnetic susceptibility, and trap-free electronic structure for Nd-doped FAPbI_3 under MF, in excellent agreement with the experimental observations of improved crystallinity, magnetic ordering, and extended carrier lifetimes. The planar PSCs fabricated under an MF achieved a PCE of $\sim 21\%$, while the Nd^{3+} -doped counterparts reached 23.8% , with the unencapsulated devices retaining $\sim 95\%$ (Nd-doped) and $\sim 68\%$ (pristine) of their initial efficiency after 900 h of continuous illumination at 50% relative humidity. These results establish the first demonstration of stable, high-performance FAPbI_3 perovskite devices -both pristine and rare-earth-doped- grown under an external MF. The excellent agreement between the DFT-predicted reduction in formation energy, enhanced magnetic susceptibility, and trap-free electronic structure, and the experimentally observed improvements in crystallinity, magnetic ordering, and carrier dynamics demonstrates the strong synergy between theoretical design and experimental realization in this work. Overall, our study demonstrates that DFT-guided dopant selection combined with MF-assisted

crystallization offers a viable strategy to achieve high-performance and durable perovskite devices through rational materials design and process optimization.

Materials and Methods

ESR measurements. Magnetic measurements were conducted using an X-band JEOL JES-FA300 electron spin resonance (ESR) spectrometer. The ESR measurements for all samples were performed at a frequency of 9.8 GHz under various MF orientations at room temperature. To investigate the angular dependence of the ESR spectra, measurements were carried out in out-of-plane geometry. Specifically, the film was systematically rotated from the surface plane to the surface normal relative to the applied DC MF.

Materials. All solvents and chemicals were obtained commercially and were used directly as received. Fluorine-doped tin oxide (FTO)-coated glass substrates (Asahi Glass, 10 Ω /sq), Neodymium trifluoromethanesulfonate ((CF₃SO₃)₃Nd), N, N-dimethylformamide (DMF), dimethyl sulfoxide (DMSO), chlorobenzene (CB), isopropanol (IPA) Acetonitrile (ACN), 4- *tert*butylpyridine (*t*BP), bis(trifluoromethylsulfonyl)imide lithium salt (Li-TFSI) were purchased from Sigma-Aldrich. 2,2',7,7'-tetrakis(*N,N*-di-*p*-methoxyphenylamine)-9,9'-spirobifluorene (spiroOMeTAD) were purchased from Dyesol. PbI₂ and methylammonium chloride (MACl) were purchased from Tokyo Chemical Industry (TCI). Formamidinium iodide (FAI) was purchased from Great Cell Solar. Tin Oxide (SnO₂) nanoparticle (15 wt% in water) was obtained from AlfaAesar Inc.

DFT calculations. We used the VASP program for plane-wave Density Functional Theory (DFT) [30,31]. In our calculations, we choose an energy cut-off of 400 eV as the basis set. We used SCAN-rVV10 functional; a meta-GGA functional SCAN and the non-local Vydrov-van Voorhis correlation (rVV10) functionals, including van der Waals interactions to describe the exchange-correlation energies

[32-34]. In our calculations, we used pseudo-potential functions to represent core electrons using the standard projector-augmented wave (PAW) method. Since our study includes the calculations of magnetic properties of materials, we needed to include spin-orbit interactions (SOC) in our DFT calculations. It is clear that SOC will be directly coupled with the crystal structure of atomic/molecular spin. Therefore, the orientation of spin magnetic moments will affect the electronic energy of the system. In the study, we included the effect of magnetization in theoretical calculations with SCAN-rVV10+SOC by restricting the spin magnetic moments to direct them to a certain crystal axis. When done in this way, it will be observed how sensitive the system is to the magnetic field (MF), more generally and advantageously, rather than how much the system is affected by any value of the MF. For SCF energy calculations, an energy threshold of 10^{-4} eV is used. The geometries are completely relaxed until all residual forces on the atoms are below 0.01 eV/Å. We used a minimum of 4x4x4 grid for k-points. Further details of the DFT calculations are provided in the SI.

Films Preparations and Device Fabrications. Perovskite solar cells (PSCs) were fabricated with the following structure: Fluorine tin oxide (FTO)/SnO₂/FAPbI₃ and (CF₃SO₃)₃Nd) doped FAPbI₃/Spiro-OMeTAD/Au. FTO substrates were cleaned sequentially in 2% Hellmanex aqueous solution, deionized (DI) water, ACN, and ethanol using sonication for 20 minutes in each solvent. The substrates were dried using compressed air and treated with UV-ozone for 20 minutes to remove residual organic contaminants. A SnO₂ ETL was deposited by diluting 0.3 mL of SnO₂ colloidal solution in 1.5 mL DI water, followed by spin-coating at 3,000 rpm for 30 seconds. The coated substrates were annealed at 180 °C for 30 minutes in ambient air. After cooling to room temperature, a secondary UV-ozone treatment was applied for 20 minutes before transferring the substrates into a nitrogen-filled glove box for perovskite deposition. The glass/FTO/SnO₂ substrates were transferred into a glove box.

Perovskite films were fabricated using a standard two-step spin-coating method with perovskite inks (<https://www.mperovskite.com/>). Firstly, a PbI₂ solution (1.5 M in DMF:DMSO, 9:1 v/v) was spin-

coated on SnO₂-coated FTO substrates at 1.500 rpm for 30 seconds, followed by annealing at 90 °C for 5 minutes. For doped samples, Nd(CF₃SO₃)₃ was added to the PbI₂ precursor at a nominal molar ratio of 0.08% relative to Pb, and the spin rate remained at 1.500 rpm. Subsequently, a solution containing organic salts (FAI:MAI:MACl in a 90:6.39:9 mg ratio dissolved in 1 mL IPA) was spin-coated at 2.000 rpm for 30 seconds and annealed at 150 °C for 10 minutes. The whole procedure was performed in a nitrogen-filled glovebox. For the MF process, a custom MF station integrated with neodymium magnets was used during the annealing process (see **Diagram S1**). The MF strength was calibrated to 100 mT using a Gauss meter, and all perovskite annealing steps were performed within this field to evaluate the effects on crystallization and morphology.

The HTM solution was prepared by dissolving 72.3 mg of spiro-OMeTAD in 1 mL of CB. To this, 28.8 µL of 4-tert-butylpyridine (tBP) and 17.5 µL of a lithium bis(trifluoromethanesulfonyl)imide (Li-TFSI) solution (520 mg/mL in ACN) were added as dopants. The prepared solution was spin-coated onto the perovskite films at 4.000 rpm for 30 seconds in a nitrogen-filled glove box to form a uniform HTM layer. The samples were then left overnight in a dry air environment to allow the oxidation of spiro-OMeTAD for optimal performance.

The Au electrode was thermally evaporated onto the spiro-OMeTAD layer under a high vacuum using a shadow mask to define the active area (typically 0.36 cm²). For the first some nm, the evaporation rate was maintained at 0.1-0.2 Å/s to ensure uniformity, and the final thickness of the gold layer was set to 100 nm.

Film and Device Characterizations. The crystallinity of perovskite films was evaluated by X-ray diffraction (XRD) (PANalytical X'Pert Pro). The top-view morphologies of the films were assessed using the field-emission scanning electron (HR-SEM) (FE-SEM, ZEISS GeminiSEM 500). The roughness of the perovskite surface was estimated with an atomic force microscope (AFM, Park System XE-100E). KPFM measurements were conducted with the same AFM system to map the surface potential

distribution. The contact angle (CA) measurements were assessed by a drop-shape analyzer (KRÜSS, DSA25) to determine the surface wettability of perovskite films. Steady-state photoluminescence (PL) and time-resolved PL (TRPL) spectra were conducted using Edinburgh Instruments (FS5). The perovskite films were excited at room temperature with a pulsed laser emitting at 405 nm to determine the carrier extraction ability. The TRPL spectra was fitted with a double-exponential decay function.

Hole-only devices were fabricated with the architecture FTO/NiO_x/perovskite (bare or Nd-doped)/spiro-OMeTAD/Au. The SCLC measurements were performed under dark conditions using a Keithley 2400 source-meter to apply a voltage bias across the device and record the corresponding current. The devices were tested in the forward bias configuration to analyze the hole transport properties within the perovskite layer. The current-voltage (I-V) data was analyzed in the Mott-Gurney framework for SCLC.

The electrochemical impedance spectroscopy (EIS) measurements were conducted in the dark to study the charge transport and recombination characteristics of the solar cells. The experiments were performed using an Ivium Compact-stat potentiostat housed in a Faraday cage to minimize external interference. A frequency range of 5 MHz to 1 Hz was explored, with devices subjected to forward bias potential at V_{OC} . Nyquist plots of the EIS data were analyzed and fitted using software to determine the equivalent circuit parameters and extract insights into the interfacial resistances and capacitances.

The photovoltaic performance of the devices was evaluated under simulated solar illumination. A 450-W Xenon light source (Oriel) was used, and the spectral mismatch with the AM 1.5G solar spectrum was calibrated using a Schott K113 Tempax filter (Prazosopms Glass & Optik GmbH). A silicon photodiode served as the reference for light intensity calibration before each measurement. The current density-voltage (J-V) characteristics were recorded using a Keithley 2400 sourcemeter with a scan rate of 50 mV/s. Typically, the devices were measured in reverse scan ($1.2\text{ V} \rightarrow 0\text{ V}$, step 0.02 V) and forward scan ($0\text{ V} \rightarrow 1.2\text{ V}$, step 0.02 V). All the devices were measured without pre-conditioning, such as light-soaking and a bias voltage. Devices were masked with a black metal aperture defining an active area of

0.09 cm². The IPCE was measured using a commercial setup (PV Measurements) equipped with a 300-W Xenon lamp as the light source. Device stability was tested under continuous illumination from a white LED lamp, and maximum power point (MPP) tracking was performed using a Fluxim system. Nonencapsulated devices were subjected to MPP tracking under varying ambient conditions to evaluate their stability.

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Author Contributions

MA, CD and IY designed and supervised the research. NŞ, FY, CD conducted the FMR experiments. NŞ, NG, PK, AC, AN, IT, KE, EÇ, SA, MA performed the synthesis, film fabrication, device measurements and all film characterizations. IY, AN, NŞ performed the DFT computations and materials screening calculations. SA, MA, IY wrote the first draft of the manuscript, and all authors contributed feedback and comments.

Competing interests

The authors declare no competing interests.

Data and materials availability

All data needed to evaluate the conclusions are present in the paper or the supplementary materials.

Supplementary Materials

Methods, Supplementary Text, Diagram S1, Figs. S1 to S22, Tables S1 to S6.

Data Availability

The data supporting this article have been included as part of the Supplementary Information.

Code Availability

The VASP software used in this article can be found at <https://www.vasp.at>. The version of the software employed for this study is version 5.4.4.

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FIGURE CAPTIONS:

Figure 1. Magnetism measurements, dopant screening procedure. **A.** A diagram showing the appearance of the magnetic easy axis via electron-spin resonance spectroscopy (ESR). **B.** ESR spectrum of a dopant-free FAPbI₃ perovskite film grown under a magnetic field. A resonance signal indicates a magnetic easy axis along a certain crystal direction. **C.** Dopant screening procedure to achieve high efficiency perovskite cells.

Figure 2. DFT calculations and computational dopant screening. **A.** Three possible positions of the dopant Xⁿ⁺ within the perovskite lattice. **B.** DFT calculated magnetization of X.FAPbI₃. **C.** Formation energies of X.FAPbI₃ under no MF and inside the MF. A Pb-rich growth condition is considered. The XPS survey spectrum of the corresponding control film (no MF) is provided in the Supplementary Information (**Figure S10**). **D.** (left) three common surface terminations, with surface dopants, we considered and their structures, (right) calculated surface energies (bars) and their average values (black line). **E.** A diagram that collectively shows the existence of a DFT-calculated trap state from the DOS structures of X.FAPbI₃ within the band-gap. **H.** A diagram to demonstrate which dopant successfully passed the computational screening process for the next step (*i.e.*, film preparation under MF and device measurements).

Figure 3. Preparation, morphology and PL spectra of perovskite films near MFs. In all figures: FAPbI₃ films prepared without (control) and with MF (control@MF), and Nd³⁺ doped FAPbI₃ films prepared with MF (Nd³⁺ 0.08% @MF). **A.** Proposed mechanism for synthesis and crystallization of FAPbI₃ perovskite with a dopant inside MF. **B.** XRD patterns of the control, control@MF, and Nd³⁺ 0.08% @MF films, highlighting the α -FAPbI₃, δ -FAPbI₃, and PbI₂ phases and the sharpening of the (100) diffraction peak under MF treatment. **C.** Steady-state photoluminescence (PL) spectra of the perovskite films on glass. **D.** Time-resolved PL spectra of perovskite films. **E.** SEM surface morphologies of prepared films.

Figure 4. PV performance and long-term operational stability of PSCs. **A.** J-V curves of the best-performing devices under both forward and reverse voltage scans: control with MF (control@MF), and Nd.FAPbI₃ with MF (Nd³⁺ 0.08% @MF). In Figures **B-F**, all three device types are included: control (no MF), control@MF, and Nd³⁺ 0.08% @MF. **B.** Steady-state PCE measurement results of the best performing PSCs. **C.** EQE spectra of the best performing PSCs. **D.** PCE distribution histogram of planar-type PSCs. **E.** Statistical results of the device parameters (10 devices in each case): V_{OC}, FF, J_{SC}. **F.** Operational stability of devices tracked at the maximum power point (MPP) (unencapsulated).

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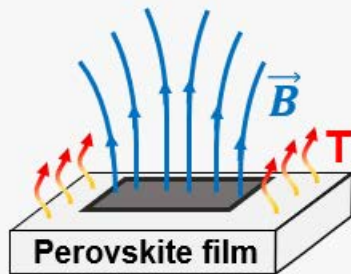
Further improvements in perovskite film synthesis remain essential for device performance and durability. Here, the authors demonstrate DFT-guided dopant screening and magnetic-field-assisted crystallization, enabling facile fabrication of highly crystalline, stable, high-efficiency formamidinium lead iodide solar cells.

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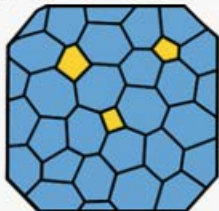
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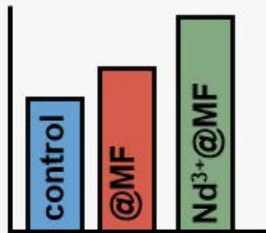


Improved
perovskite films

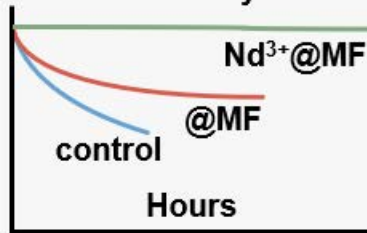


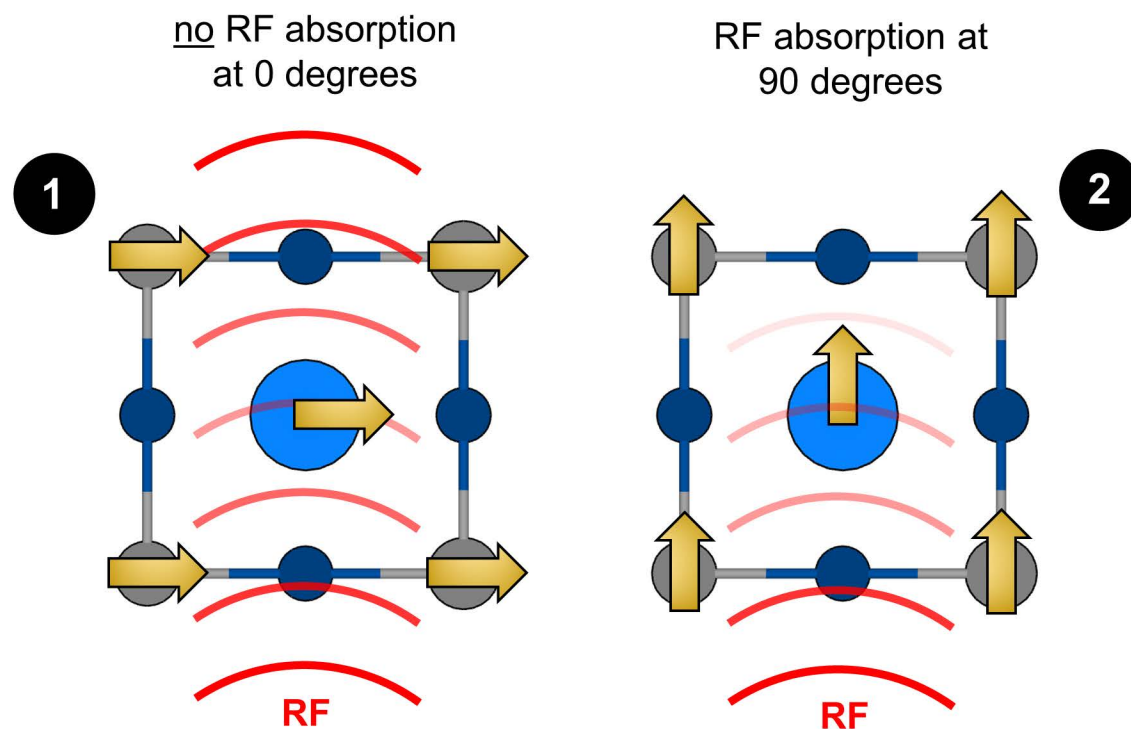
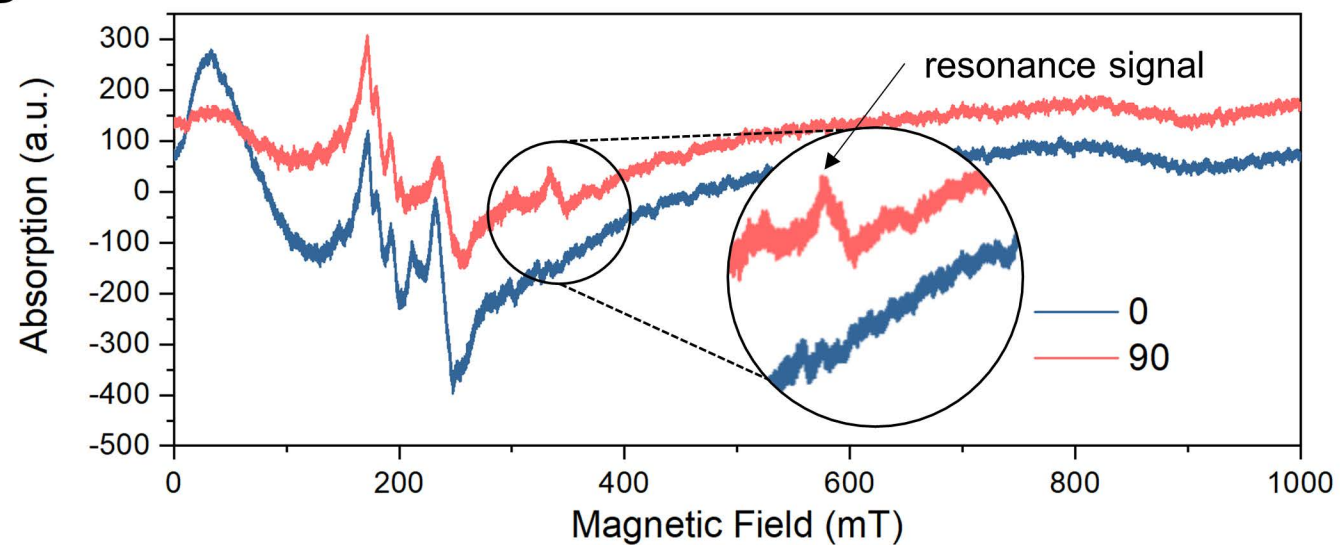
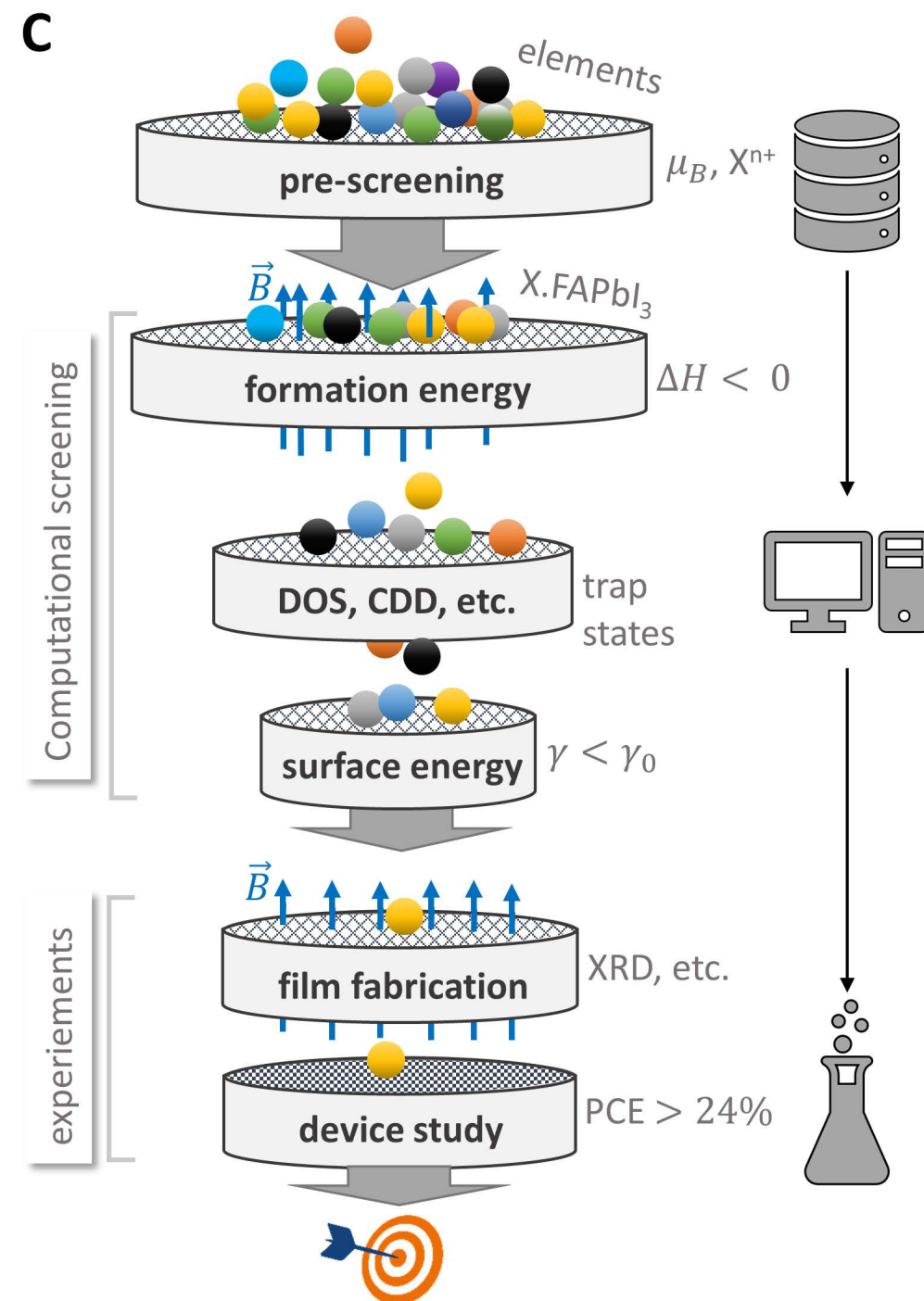
Low defects

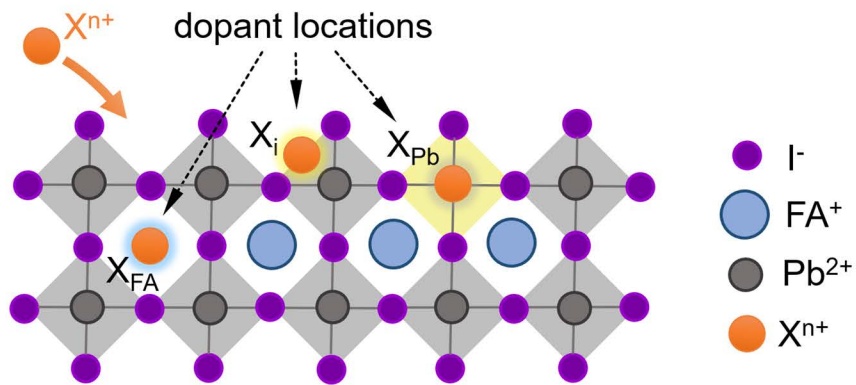
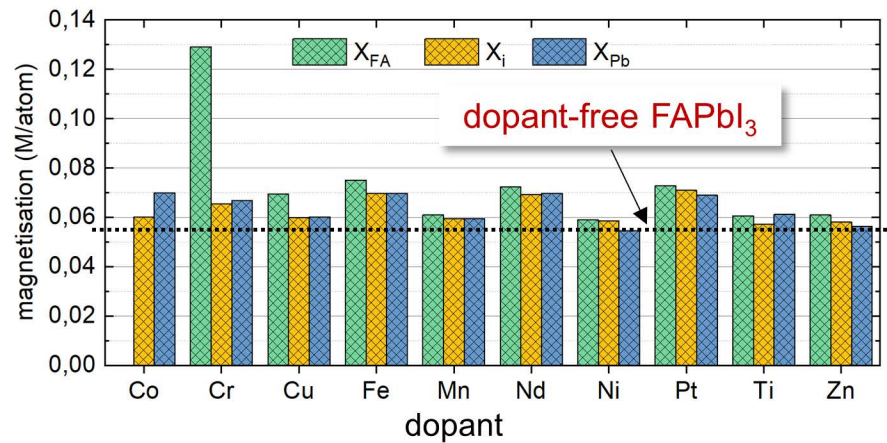
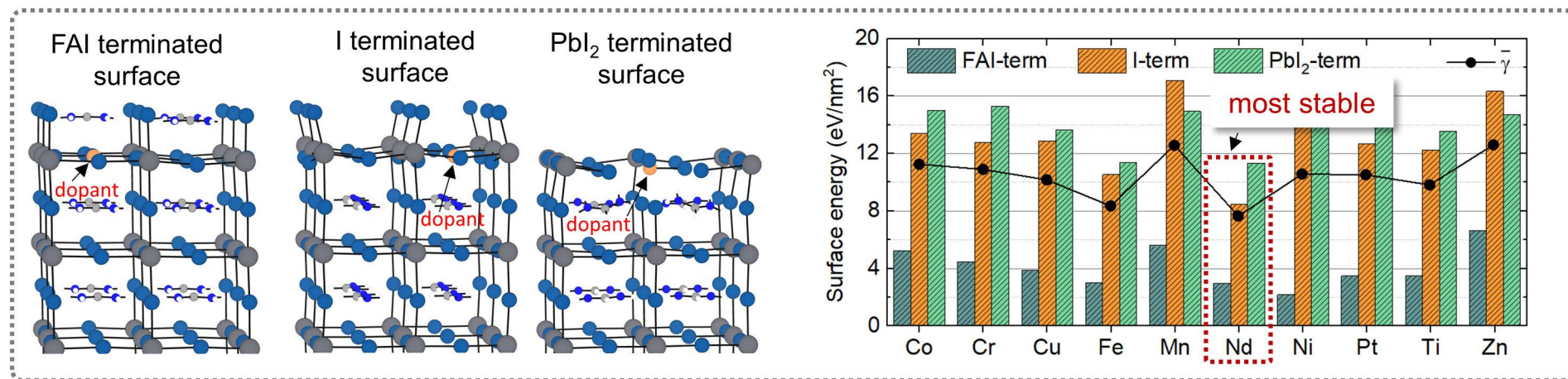
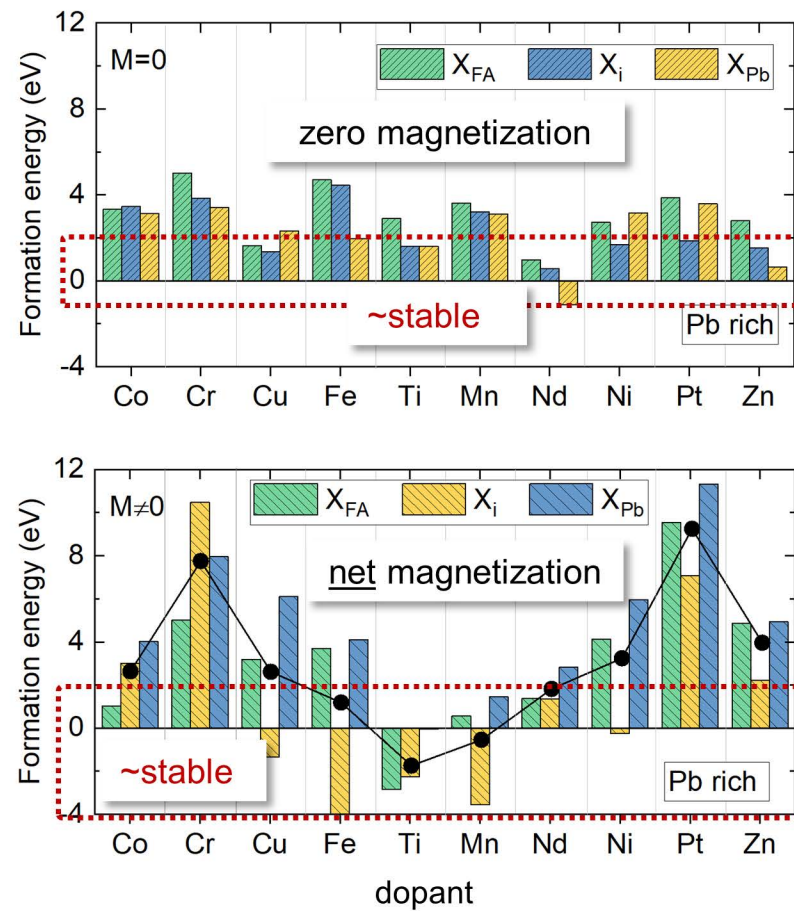
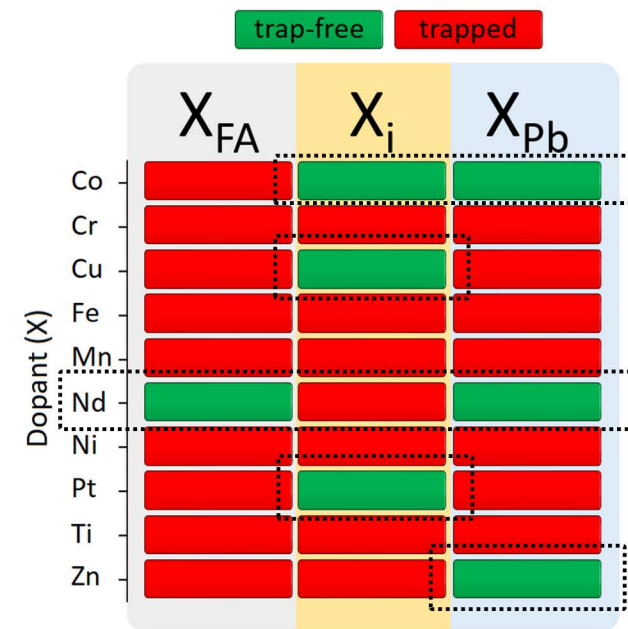
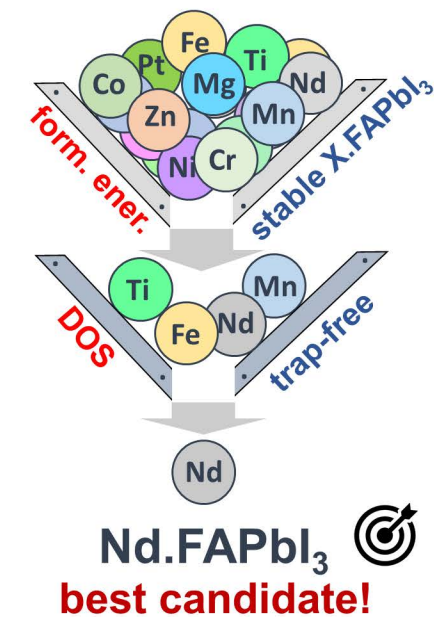
PCE

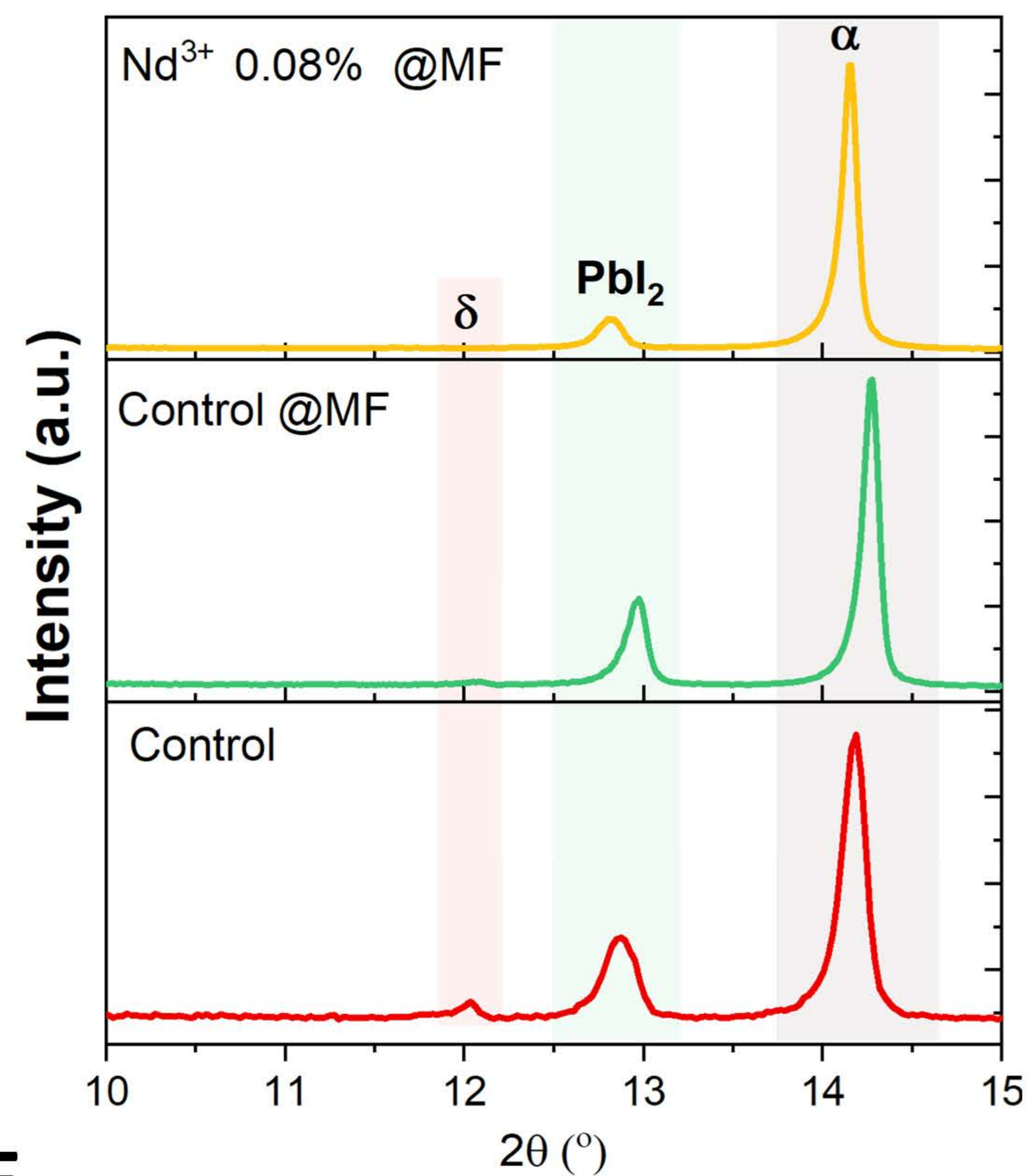
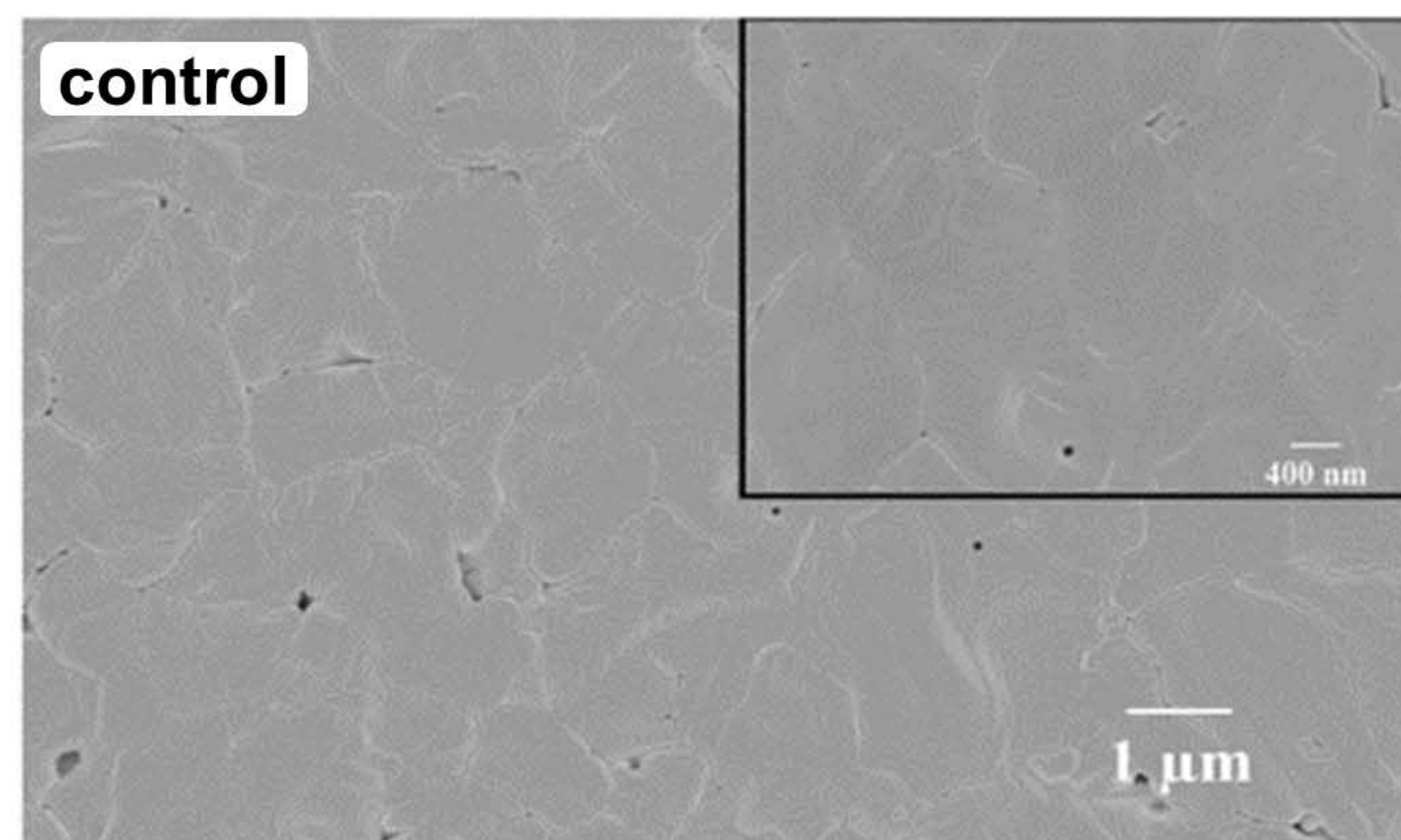
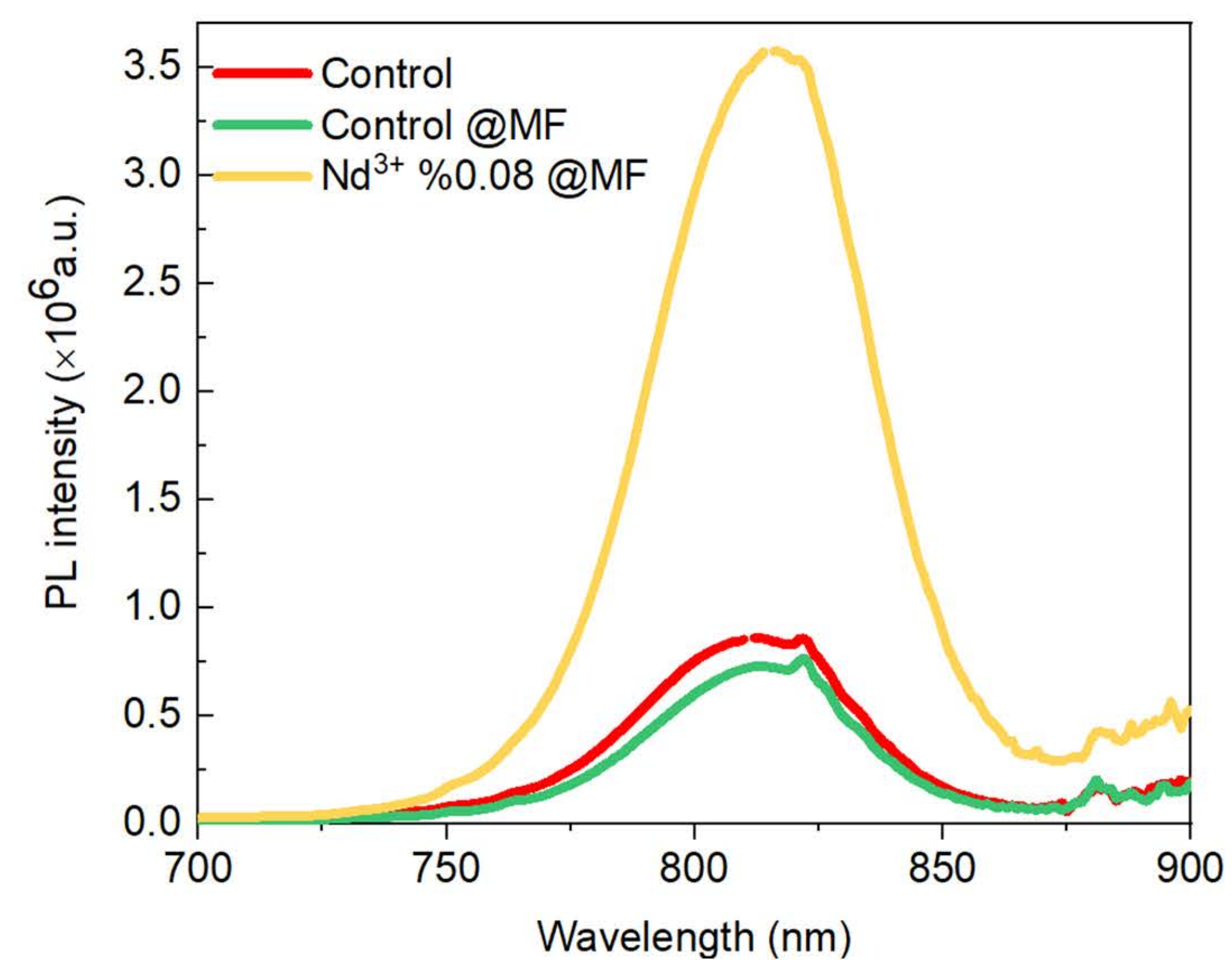
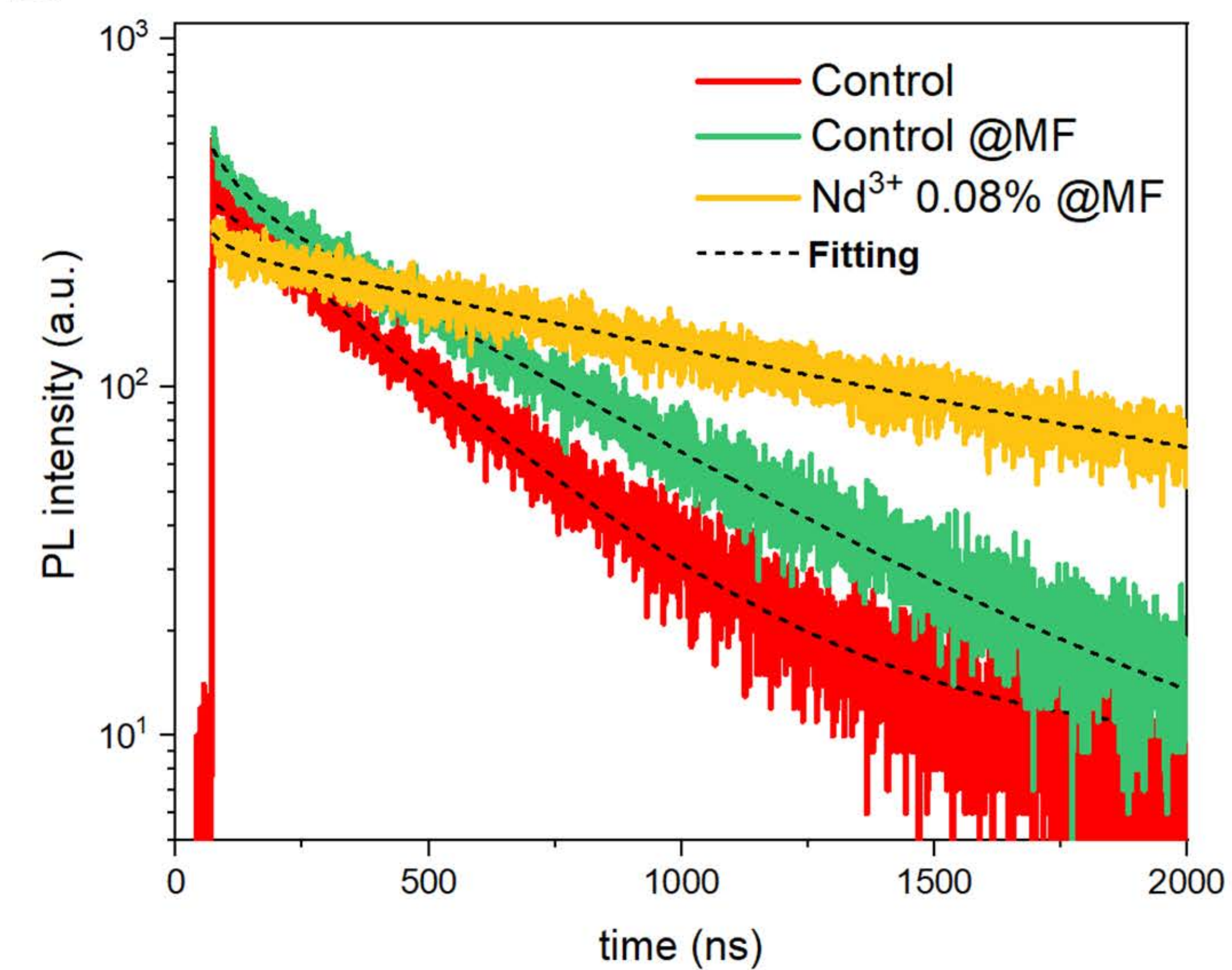


Stability



A**B****C**

A**B****D****C****E****F**

B**E****C****D****A**

FA(Pb:X)I_3 /
DMF:DMSO

